the OTA Requestor and details for available Software Images, and obtains a consent signal based on an Administrator having previously stated an account preference to "Always apply software updates to light bulbs". The OTA Provider then proceeds further.

User consent delegation to Nodes

Some capable Nodes MAY have sufficient hardware capabilities to request user consent by means such as display or voice, and subsequently recover user consent feedback through input mechanisms. These devices MAY request optional delegation of user consent by the following method:

1. The OTA Requestor SHALL set the RequestorCanConsent field in the QueryImageRequest to True, indicating ability to obtain consent.
2. The OTA Provider, if it determines that the best way to obtain user consent is to delegate to the OTA Requestor Node, SHALL include the UserConsentNeeded field, with a value set to True in the QueryImageResponse, indicating that user consent was not previously obtained, and that delegation SHALL occur.
3. The OTA Requestor, upon observing presence of UserConsentNeeded field set to True and the availability of an image in the QueryImageResponse SHOULD proceed to obtain user consent using its onboard means, prior to transferring the OTA Software Image reported. If the UserConsentNeeded field is set to False or absent, the OTA Requestor SHALL assume that the OTA Provider already obtained sufficient user consent during the querying phase.

The above method of obtaining User Consent at the OTA Requestor level SHALL NOT be used if a Node is configured with the LocalConfigDisabled attribute set to True as reflected by the Basic Information Cluster.

User consent recommendations

Because of the variety of Vendors and Devices, the concept of "User Consent" will necessarily take many different forms. Therefore, it is RECOMMENDED that every implementer of OTA Provider logic implement a transparent and easy-to-use set of functionality to allow Users to provide or deny consent for software updates, in a way that functionally integrates with their products and respects the general requirements stated above. Implementation of this feature is expected to improve "user experience" and assist with building trust regarding the installation of new software on Nodes.

It is RECOMMENDED that any method of consent that stores consent signals also provide a way to revoke this consent in the future.

It is RECOMMENDED that metadata from Software Images be used to convey as much information as required within the available set, so that a User can make an informed decision based on the nature of the product being updated, the human-readable instance of the new version number (e.g. SoftwareVersionString in OTA Image Header), the changes made available, and their side-effects on product functionality. Any URL for online contents conveyed during this process SHOULD point to content that can be localized at the time of delivery, whenever possible. The responsibility for the maintenance of such version information is on the Vendor providing the URL and metadata. The OTA Provider and associated implementation-specific logic SHALL allow a User to consent to an update, even if errors occur while trying to provide additional release information, as the metadata within the Software Image should suffice to provide a first-order description of the new version, which could then be researched or cross-referenced by the User.

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